

Lane Change Overtake Assistance System Using LoRa Based V2V Communication

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This project presents a Lane Change Assistance System to improve road safety and reduce accidents. Unsafe lane changing is one of the major causes of road accidents in modern traffic conditions. Drivers often fail to detect nearby vehicles while changing lanes. To overcome this issue, an intelligent system combining image processing and communication is proposed. The system uses a camera to capture real-time road images. Image processing techniques are applied to detect lane boundaries accurately. This helps in identifying whether the vehicle is within its lane or attempting to change lanes. In addition, Vehicle-to-Vehicle (V2V) communication is implemented using LoRa technology. The LoRa module enables long-range communication between vehicles. When a vehicle intends to change lanes, a signal is transmitted to nearby vehicles. The receiving vehicle processes the signal and generates a warning alert. This helps drivers become aware of surrounding vehicles in real time. The combined approach improves the reliability compared to single-method systems. It overcomes the limitations of image processing in poor visibility conditions. It also enhances communication-based systems by adding lane detection capability. The system is cost-effective and suitable for real-time implementation. Overall, this project contributes to safer and smarter transportation systems.

Keywords: Lane Detection; Image Processing; V2V Communication; LoRa Technology; Road Safety.

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Abbreviations

Abbreviation	Description
V2V	Vehicle-to-Vehicle Communication
LoRa	Long Range Communication Technology
ITS	Intelligent Transportation System
ADAS	Advanced Driver Assistance System
GPS	Global Positioning System
RF	Radio Frequency
CPU	Central Processing Unit
GPU	Graphics Processing Unit
UART	Universal Asynchronous Receiver Transmitter
SPI	Serial Peripheral Interface
I2C	Inter-Integrated Circuit
DSP	Digital Signal Processing

CHAPTER 1

Introduction

1.1 Introduction

Road safety has been a major concern in recent times due to the rapid increase in the number of vehicles on the road. Among the major causes of accidents, lane changes and overtaking have been prominent. The major causes of such accidents are blind spots, lack of awareness among drivers, and delayed reactions. Driver assistance systems have been in use for some time. However, these systems have some drawbacks in terms of accuracy. Thus, there is a need for more advanced systems that can assist in lane changes and overtaking. The present project aims at designing a Lane Change and Overtake Assistance System by implementing Vehicle-to-Vehicle (V2V) communication technology. The system can assist in improving road safety by communicating between vehicles over a large distance with less power consumption. The technology that can support such a communication is the Long Range technology, also known as LoRa technology. The technology has the potential for wireless communication over a large distance by consuming less power. The proposed system not only incorporates the use of LoRa technology for vehicle communication but also incorporates image processing techniques for lane detection. Moreover, Digital Signal Processing techniques have been employed in order to increase the quality of the signal sent. The importance of this project is that it contributes to the intelligent transportation system and can be used in the future for autonomous vehicles. The system is important because it can provide an efficient solution for improving road safety, especially in cases where the existing system is not effective. The main purpose of this project is to create an efficient system that can effectively detect the surroundings, including the lane positions, and provide the required information to the driver. The scope of the project includes creating the

hardware and software components, developing the communication protocols, and testing the system. The project is divided into the following chapters: Chapter 1 is the introduction to the project. Chapter 2 is the literature review of the existing technologies. Chapter 3 is the system design, including the methodology. Chapter 4 is the implementation part. Chapter 5 is the results. Chapter 6 is the conclusion.

1.2 Background and Motivation

Intelligent transportation systems (ITS) is one of the important research areas that gained significant attention in recent times with the rapid growth in the number of vehicles on the road and the demand for safe and efficient road transportation. Today, vehicles are being equipped with advanced driver assistance systems (ADAS), which help in reducing the chances of accidents and making the driving experience much safer. Of all the driving activities, changing lanes and overtaking other vehicles are considered to be the most critical, as these involve interactions with many vehicles on the road. In the context of road safety, the technology used to ensure safety on the road has evolved over time from simple warning-based safety to sensor-based safety. Initially, the safety technology used on the road involved simple indicator-based safety and mirrors. Later, technologies such as radar, ultrasonic sensors, and cameras were introduced to sense the presence of vehicles on the road. Even though these safety technologies improved the safety standards on the road, they still remain limited in terms of the distance they can sense, environmental factors such as fog and rain, and the inability to communicate with other vehicles. Recently, with the advent of advanced wireless communication technology, vehicle-to-vehicle communication technology has emerged. Using this technology, vehicles can communicate with each other and exchange data such as vehicle speeds, positions, and directions. Technologies such as Dedicated Short Range Communication (DSRC) and Wi-Fi have been studied extensively for implementing this technology. Even though these technologies can be used for implementing vehicle-to-vehicle communication technology, they still

remain limited in terms of the power consumption and the distance over which the communication can take place. LoRa, also known as Long Range, has also been identified as a potential solution for V2V communication because of its ability to enable communication over a long range with low power consumption. LoRa technology has already been successful in various Internet of Things-based applications and has shown reliable performance in various wireless communication systems. Thus, LoRa technology has a great potential to be integrated with vehicular communication systems to improve the range of information communication. At the same time, various image processing techniques have also been widely used to improve lane detection and vehicle tracking systems. Lane Departure Warning Systems (LDWS) have also been widely discussed, where cameras are used to detect lanes and assist drivers in maintaining appropriate lanes. However, these systems operate independently and do not include communication with other vehicles. Thus, despite various technologies being identified for lane change detection and communication, a gap still exists in integrating these technologies to enable long-range communication along with real-time lane detection systems. This has led to a need to design a system that can address various issues in lane change detection and communication. The motivation behind undertaking this project is based on the increased number of accidents on the road, where unsafe lane changes and overtaking maneuvers are a major concern. Thus, a strong need exists to design an intelligent system that can assist in providing early warnings to prevent such accidents. This project seeks to address these challenges through the development of a Lane Change and Overtake Assistance System using LoRa-based V2V communication with the incorporation of image processing and Digital Signal Processing techniques. It is believed that the proposed system is not only beneficial to the field of vehicle communication, but it is also beneficial in the field of intelligent transportation systems.

1.3 Problem Statement

Road safety remains a critical concern worldwide, with a significant number of accidents occurring during lane changing and overtaking maneuvers. These accidents are primarily caused by human errors such as lack of awareness, delayed reaction, and inability to accurately judge the position and speed of nearby vehicles. Blind spots and high traffic density further increase the risk of collisions, making lane change decisions highly complex and error-prone. Existing driver assistance systems mainly rely on sensors such as cameras, radar, and ultrasonic devices to detect nearby vehicles. While these systems provide some level of support, they are limited by factors such as restricted detection range, environmental conditions (e.g., fog, rain, and low visibility), and lack of direct communication between vehicles. As a result, they fail to provide complete situational awareness, especially in high-speed or congested traffic scenarios. Moreover, most current systems operate independently within a single vehicle and do not utilize real-time Vehicle-to-Vehicle (V2V) communication. This absence of communication leads to delayed or insufficient information about surrounding vehicles, reducing the effectiveness of safety systems. Additionally, existing solutions often lack integration with lane detection mechanisms, which are essential for understanding the vehicle's position on the road. Therefore, the core problem addressed in this project is the lack of a reliable and integrated system that can provide real-time information about surrounding vehicles and assist drivers during lane change and overtaking maneuvers. There is a need for a system that combines long-range communication, accurate lane detection, and real-time alert generation to enhance driver awareness and reduce the likelihood of accidents. This project aims to address these challenges by developing a LoRa-based Vehicle-to-Vehicle (V2V) communication system integrated with image processing and Digital Signal Processing (DSP) techniques. The proposed system focuses on improving communication range, reliability, and situational awareness, thereby providing a practical and efficient solution to enhance road safety.

1.4 Objectives of the Project Work

This section of the report explains the objectives of the proposed Lane Change and Overtake Assistance System by using the Vehicle-to-Vehicle communication technology through the LoRa protocol. The objectives of the system have been framed in such a way that they can fulfill the shortcomings of the current technology, which is being used for Vehicle-to-Vehicle communication. The objectives of the project work are:

- Develop a LoRa-based V2V communication system.
- Implement lane detection using image processing techniques.
- Integrate sensors for vehicle detection and monitoring.
- Apply Digital Signal Processing (DSP) for signal processing.
- Design a real-time alert system for driver assistance.
- Evaluate the overall system performance.

1.5 Organization of the Report

This report has been organized in such a way that each chapter of the report deals with different aspects of the project work. The report has been organized in the following manner:

- **Chapter 2 : Literature Review** – In this chapter of the report, the literature related to the research work has been discussed in detail.
- **Chapter 3: System Design** – In this chapter of the report, the design of the system has been explained in detail.
- **Chapter 4: Implementation** – In this chapter of the report, the implementation of the system has been explained in detail.
- **Chapter 5: Results and Discussion** – In this chapter of the report, the results of the system have been explained in detail.

- **Chapter 6: Conclusion and Future Scope** – In this chapter of the report, the conclusion of the system has been explained in detail.

CHAPTER 2

Literature Survey

2.1 Introduction

The literature survey is of significant importance in understanding the existing research and technological developments regarding the proposed project. The literature survey aims to provide a comprehensive survey of existing research work in the domains of Vehicle-to-Vehicle (V2V) communication, LoRa technology, lane detection systems, and driver assistance mechanisms. Through the existing research work, it is possible to identify the pros and cons of the existing solutions. In recent times, a significant volume of research work has been conducted in the domain of intelligent transportation systems (ITS) for improving road safety and driving efficiency. Different methodologies have been explored for assisting drivers during critical driving maneuvers such as lane changing and overtaking. However, most of the methodologies focus only on a limited aspect of the problem and do not provide a comprehensive solution. The methodology used for this literature survey is based on searching and selecting research papers from IEEE publications, conferences, and other reliable sources. The papers are selected based on their contributions to key areas such as LoRa-based communication, V2V and V2X communication, lane detection algorithms, and accident analysis. The objective of this literature survey is to assess the existing system based on its performance in terms of communication, reliability, real-time capability, and safety features. Another objective of this literature survey is to identify the research gap, which will help in developing a new system. It is identified that although LoRa technology helps in long-range communication, image processing helps in accurate lane detection, there is a lack of research in this area, which combines both technologies for a unified system. This chapter is divided as follows: first, this chapter discusses the research papers on LoRa-based communication, then

moves on to discuss papers on V2V and V2X communication, followed by papers on lane detection and accident analysis, and finally concludes with the limitations of existing systems and identifies the research gap, which is addressed in this work.

2.2 Review of Prior Research

In the last few years, extensive research has been conducted in the domains of Vehicle-to-Vehicle (V2V) communication, wireless technologies, and driver assistance systems to ensure safety on the roads. This section of the report focuses on the review of some of the most important research works related to the proposed project. A. P. A. Torres, et al. in their research work titled “LoRa Technology in Vehicle Communication: An Experimental Study” conducted an experimental study to assess the performance of LoRa technology in terms of communication range, reliability, and packet delivery ratio in a vehicular environment. The results showed that LoRa technology can be used to provide long-range communication with low power consumption, which can be used in V2V communication. However, the study only focused on the performance of LoRa in terms of communication and did not consider the application of LoRa in safety-related functions, such as lane change assistance. A. J. Wixted, et al. in their research work titled “LoRa and LoRaWAN in Wireless Sensor Networks: Performance Evaluation” analyzed the performance of LoRa and LoRaWAN technologies in wireless sensor networks. The results showed that LoRa technology can be used to provide long-distance communication with low power consumption. Although the study provided important insights into wireless communication, it only focused on wireless sensor networks and did not consider the performance of LoRa in a vehicular environment. Y. Li et al. (2018) assessed the performance of LoRa technology in V2X communication scenarios. The performance assessment considered the effects of vehicle speed, vehicle mobility, and distance. The results showed that the performance is affected by the mobility of the vehicle. However, the application of LoRa technology in the design of driver assistance systems is not

addressed. M. Shawky (2020) investigated the factors that cause lane change accidents. The investigation results showed that driver errors, blind spots, and lack of situational awareness contribute to accidents. It is evident that the importance of designing advanced driver assistance systems to reduce the chances of accidents is highlighted. However, the application of the technology in designing the driver assistance system is not addressed. K. C. Kavya and M. A. Anusuya (2019) designed a lane detection system using image processing techniques to design the lane departure warning system. The results showed the success of the designed system. It is evident that the designed system can be applied in the design of the driver assistance system. However, the designed system does not support communication with other vehicles. F. Re et al. (2021) evaluated the performance and reliability of commercial lane departure warning systems. The performance assessment considered the performance of the system under different driving scenarios. It is evident that the importance of designing improved systems to operate effectively in different environments is highlighted. From the above analysis, it is evident that both communication technology and the design of the lane detection system have been well addressed. However, the application of the technology in the design of the driver assistance system is not addressed.

2.3 Identified Research Gaps

After conducting the required research on the prior work done in the field, several limitations and gaps in the existing systems have been identified, which can be listed as follows:

- Most of the existing systems either address the communication technologies or the lane detection techniques, but the integration of both into one single system is not addressed.
- The existing V2V communication technologies like DSRC and Wi-Fi also have limitations in terms of the distance, power consumption, and reliability for long-distance communication.

- Although the LoRa-based communication technology is studied for its long distance communication ability, its application in the field of real-time driver assistance is still limited.
- The image processing-based lane detection techniques also have limitations in terms of environmental conditions such as low illumination, fog, and rain.
- The existing techniques lack the feature of real-time data exchange between vehicles, which is required to enhance the awareness among the drivers.
- Very little research is done on the integration of Digital Signal Processing techniques to improve the reliability in the field of vehicular communication.

Therefore, the proposed project is designed to address the above limitations by integrating the LoRa-based V2V communication technology with the image processing and digital signal processing techniques to design an efficient and reliable lane change and overtaking assistance system.

2.4 Summary

This chapter has provided an extensive review of existing research related to vehicular communication systems, driver assistance technologies, and lane detection mechanisms. Various existing research works related to LoRa-based communication technologies, Vehicle-to-Vehicle (V2V) and Vehicle-to-Everything (V2X) technologies, image processing techniques, and accident analysis were reviewed. From the existing research works, it can be analyzed that LoRa-based communication technologies have shown promising results in terms of advantages, including long-range communication, low power consumption, and cost-effectiveness, which can be considered for vehicular communication systems. Similarly, V2V communication plays an important role in enhancing the safety of vehicles. In addition, image processing-based lane detection systems have shown promising results in terms of lane detection. However, there are

many limitations in existing research works related to vehicular communication systems. Most of the existing works have focused only on individual aspects, not considering the overall aspects. Communication technologies have shown limitations in terms of latency and reliability, while image processing-based technologies have shown limitations in terms of environmental issues. The identified gaps in the research reveal the need for an integrated solution that can efficiently communicate, process information in real-time, and utilize intelligent decision-making capabilities. Closing these gaps is vital in order to increase road safety and minimize the risk of accidents during key vehicle maneuvers such as lane changes and overtaking. Hence, the proposed project aims to design a comprehensive solution that incorporates LoRa-based V2V communication, image processing, and Digital Signal Processing techniques. It is assumed that such a solution can increase the reliability of communication, enhance situational awareness, and provide effective assistance in real-time scenarios. Finally, in conclusion, this literature survey has provided a strong base for the proposed research by identifying the challenges in the current state of the art and the need for an integrated solution. The next chapter of this report will discuss the system design and methodology of the proposed solution.

CHAPTER 3

Analysis of Current Systems and Research Techniques

The analysis of current methods and systems for driver assistance and vehicular communication is covered in this chapter. The goal of this chapter is to give readers a clear understanding of the performance, limitations, and implementation of current techniques. Numerous strategies have been put forth to increase driving efficiency and road safety since the introduction of intelligent transportation systems. These methods include image processing-based lane detection, wireless sensor networks, vehicle-to-vehicle communication, and driver assistance systems. All of these methods help to increase traffic safety and lower the likelihood of collisions. Vehicle communication systems already use current communication technologies like cellular networks, Wi-Fi, and dedicated short range communication. Although these technologies yield respectable results, they have certain drawbacks, including short range, high power consumption, and network congestion in densely populated areas. New technologies like LoRa (Long Range Communication), which offer long-distance communication with low power consumption. Additionally, image processing techniques are used to implement driver assistance mechanisms like lane detection. Although the image processing-based methods yield respectable results, environmental factors have a significant impact on their effectiveness. In order to improve current methods and give a clear understanding of how these methods can be combined to produce a dependable solution, this chapter covers a technical analysis of current methods, their benefits, and drawbacks. In order to enhance system performance and dependability, a new system incorporating LoRa-based communication, image processing, and DSP techniques will be implemented with the aid of the knowledge gathered from this analysis. The existing communication technology, such as Dedicated Short Range Communication, Wi-Fi, and cellular networks, is already implemented in the vehicular communication

system. It provides reasonable results, but it is also associated with some limitations. To overcome the limitations, new technologies such as LoRa (Long Range Communication), which provides communication over long distance with minimal power consumption, can be employed. In addition, driver assistance techniques, such as lane detection, can be employed using image processing techniques, which provide reasonable results, but the performance is highly dependent on the environmental conditions. This chapter provides the technical analysis of the existing techniques, advantages, and disadvantages, which can be helpful in enhancing the performance of the existing techniques. It provides a clear idea about the techniques to be integrated to obtain the desired results. This knowledge can be helpful in implementing the new system, which can incorporate the LoRa communication, image processing techniques, and the application of DSP techniques to enhance the performance.

3.1 Analysis of Existing Communication Systems

The existing communication techniques in the vehicular communication system play a vital role in enhancing the performance. Various communication techniques, such as Dedicated Short Range Communication, Wi-Fi, and cellular networks, are employed for vehicle-to-vehicle communication. Dedicated Short Range Communication is designed to support automotive applications, which provides low latency communication. But it is also associated with some limitations, such as the communication distance is limited. Similarly, the application of the Wi-Fi communication technique provides high data rates, which is not suitable for long distance communication because the signals are highly prone to attenuation and interference. Cellular networks can be employed for communication, which provides long distance communication, but it is also associated with some limitations, such as the introduction of latency. To overcome these problems, a new technology known as LoRa (Long Range communication) has been introduced. It provides communication over a large distance with less power consumption, which is beneficial for vehicular networks. However, its integration with the real-time system of the driver

assistance system is still in the research phase.

3.1.1 Performance Comparison of Communication Technologies

The following table represents the performance comparison of different communication technologies used in vehicular networks:

Table 3.1: Comparison of Communication Technologies

Technology	Range	Power Consumption	Limitation
DSRC	Short	Medium	Limited coverage
Wi-Fi	Medium	High	Signal interference
Cellular	Long	High	Network dependency
LoRa	Very Long	Low	Low data rate

3.2 Analysis of Lane Detection Systems

Lane detection is a significant aspect of the vehicle assistance system. Lane detection technology utilizes cameras to recognize lane markings on the road. Based on the lane detection technology, the vehicle can be kept straight. The techniques used in lane detection technology are: Edge detection Hough Transform Machine learning The techniques mentioned above help in maintaining the lane of the vehicle. However, the performance of these techniques is affected by environmental factors such as poor light, rain, fog, and road conditions. Thus, lane detection technology is not reliable in such cases.

3.2.1 Mathematical Model for Signal Processing

The Digital Signal Processing (DSP) techniques are used for signal processing. The representation of signal processing can be expressed as:

$$y(n) = x(n) + w(n) \quad (3.1)$$

where $x(n)$ is the original signal, $w(n)$ is the noise, and $y(n)$ is the received signal. Filtering techniques are used in these systems in order to remove the noise and increase the accuracy of the received data.

3.3 Integration Challenges in Existing Systems

One of the major challenges in existing systems is the lack of integration between communication and sensing technology. Most of these systems are independent and focus either on communication or sensing technology. Some of the key challenges in these systems are:

- Lack of real-time coordination between vehicles
- Limited communication range in these systems
- High power consumption in wireless technology
- Reduced accuracy in adverse environmental conditions

These are some of the key challenges in existing systems, and a unified system is required in order to address these challenges.

3.4 Illustration of System Components

3.4.1 Block Representation

The above figure represents how different components in a vehicular communication system interact with each other.

3.5 Summary

This chapter has given a detailed analysis of different systems used in vehicular communication systems and driver assistance systems. Various communication techniques were studied in detail, including DSRC technology, Wi-Fi technology, cellular technology, and LoRa technology. It has been observed that these systems are reliable in terms of communication; however,

Comparison of Communication Technologies

	 DSRC	 Wi-Fi	 Cellular	 LoRa
Range	Short	Medium	Long	Very Long
Power Consumption	Medium	High	High	Low
Limitation	Limited Coverage	Signal Interference	Network Dependency	Low Data Rate

Figure 3.1: Basic Representation of Vehicular Communication System

there are certain limitations associated with these systems. In addition, the lane detection system that makes use of image processing techniques was also analyzed. The lane detection system is an important aspect that helps the driver while driving on the road by detecting the lane boundaries and ensuring proper alignment of the vehicle. The system is highly dependent on the environmental conditions such as lighting, weather, and road quality. In addition, the study also emphasized the importance of Digital Signal Processing techniques in improving the quality of the signal while ensuring less noise in the communication system. The accuracy of the signal is important while dealing with communication systems. Moreover, the analysis also revealed that the existing systems are not properly integrated with the communication system, as they are dealing with the detection system individually. This reduces the efficiency of the system as the existing systems are not properly coordinated with the communication system. The lack of coordination among the vehicles also increases the probability of accidents while changing lanes or while overtaking. There's an urgent need for a smarter system that unites vehicles seamlessly.

CHAPTER 4

System Design and Implementation

4.1 Introduction

This chapter explains the detailed design and implementation of the proposed Lane Change Decision Aid and Warning System by utilizing Vehicle-to-Vehicle (V2V) communication technology through the use of LoRa. The system has been designed to improve road safety by promoting Vehicle-to-Vehicle communication. The system will help in improving road safety by assisting in lane change decisions for the vehicle. Lane change is one of the most critical maneuvers in vehicle control. The maneuver is more challenging in high-speed roads such as highways. Most of the accidents happen due to blind spots, distractions, and lack of information about other vehicles. The system has been designed to address these issues by utilizing communication technology. The system will help in improving road safety by promoting Vehicle-to-Vehicle communication. The system will help in improving road safety by assisting in lane change decisions for the vehicle. Lane change is one of the most critical maneuvers in vehicle control. The maneuver is more challenging in high-speed roads such as highways. Most of the accidents happen due to blind spots, distractions, and lack of information about other vehicles. The system has been designed to address these issues by utilizing communication technology. The system will help in improving road safety by promoting Vehicle-to-Vehicle communication. The system will help in improving road safety by assisting in lane change decisions for the vehicle. Lane change is one of the most critical maneuvers in vehicle control. The maneuver is more challenging in high-speed roads such as highways. Most of the accidents happen due to blind spots, distractions, and lack of information about other vehicles. The system has been designed to address these issues by utilizing communication technology. The system will help in improving road safety by promoting Vehicle-to-Vehicle communication.

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4.1.1 GPS Module

The GPS module is used to obtain the geographical coordinates of the vehicle in real time and the speed of the vehicle. This information is required to calculate the distance between the vehicles.

4.1.2 Ultrasonic Sensor

The ultrasonic sensor is used to find the distance between the two vehicles. The ultrasonic sensor increases the accuracy of the system for short distances between the vehicles.

4.1.3 Warning System

The warning system includes LED indicators and a buzzer. The warning system will warn the driver of any unsafe conditions detected by the system.

4.2 Software Requirements

The following software tools are required for the development of the project:

- Arduino IDE for coding and uploading the code
- Embedded C language for coding
- Serial communication tools for debugging the code

4.3 Working Principle

This is a continuous process running in real time. The steps involved in the working process are as follows:

1. GPS module is used to collect location and speed data
2. Ultrasonic sensor is used to collect data on the distance
3. Data is processed using the microcontroller
4. Data is transmitted to other vehicles using the LoRa module
5. Data is received from other vehicles
6. Calculation of distance and relative speed
7. Calculation of safety conditions
8. Calculation of the warning

This process repeats continuously to ensure updated information.

4.4 Data Communication Format

The transmitted data follows a structured format:

<ID, Speed, Latitude, Longitude, Direction>

This structured format ensures proper decoding and efficient communication between vehicles.

4.5 Mathematical Model

4.5.1 Distance Calculation

The distance between two vehicles is calculated using the Euclidean distance formula:

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (4.1)$$

4.5.2 Relative Speed

$$V_r = V_2 - V_1 \quad (4.2)$$

4.5.3 Safe Distance Condition

$$D > D_{safe} \quad (4.3)$$

If the distance is less than the safe distance, the system considers it unsafe and generates a warning.

4.6 Decision Logic

The decision-making process is based on distance and relative speed. If a vehicle is too close or approaching at a high speed, the system alerts the driver.

The logic ensures that the driver receives warnings only when necessary, reducing false alerts.

4.7 Block Diagram

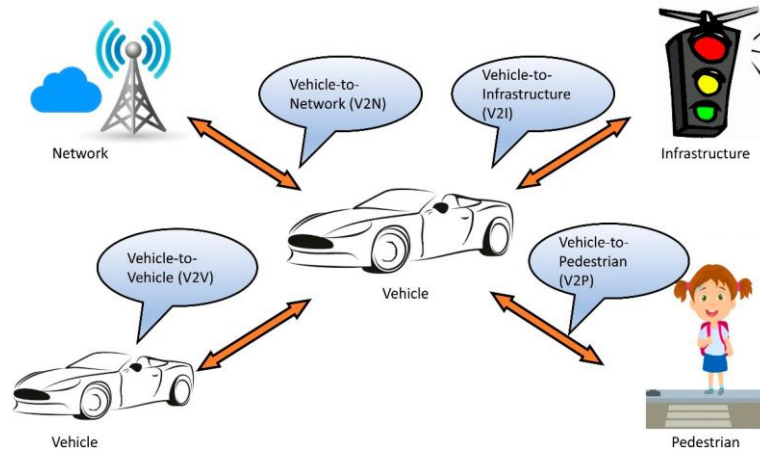


Figure 4.1: Block Diagram of Proposed System

The proposed system is based on Vehicle-to-Everything (V2X) communication using LoRa technology. The block diagram illustrates the interaction between different entities such as vehicles, infrastructure, network, and pedestrians to ensure safe and efficient lane changing.

The system mainly consists of the following components:

- **Vehicle Unit:** Each vehicle is equipped with sensors, GPS module, and a LoRa communication module. It continuously monitors parameters such as speed, position, and direction.
- **Vehicle-to-Vehicle (V2V) Communication:** Vehicles communicate with nearby vehicles to exchange real-time information like speed, distance, and lane position. This helps in avoiding collisions and assists in safe lane changing.
- **Vehicle-to-Infrastructure (V2I) Communication:** The vehicle interacts with road infrastructure such as traffic signals. This provides information about traffic conditions, signal timings, and road status.
- **Vehicle-to-Network (V2N) Communication:** The system connects to a central network or cloud through LoRa communication. This enables data sharing, remote monitoring, and advanced analysis.

- **Vehicle-to-Pedestrian (V2P) Communication:** The system detects pedestrians and alerts the driver to prevent accidents, especially in urban environments.
- **LoRa Communication Module:** LoRa technology is used for long-range, low-power communication between vehicles and other entities. It ensures reliable data transmission even over large distances.

The overall system works by collecting real-time data from sensors and GPS, transmitting it using LoRa, and receiving data from nearby vehicles and infrastructure. Based on this information, the system analyzes the surroundings and provides warnings or assistance to the driver for safe lane changing.

4.8 Algorithm

1. Initialize all hardware components
2. Read data from GPS and sensors
3. Transmit data using LoRa
4. Receive data from nearby vehicles
5. Calculate distance and relative speed
6. Compare values with safety threshold
7. If unsafe:
 - Turn ON LED
 - Activate buzzer
8. Else:
 - Indicate safe condition
9. Repeat continuously

4.9 Sample Data Analysis

Table 4.1: Vehicle Data Analysis

Vehicle ID	Speed (km/h)	Distance (m)	Status
V1	60	15	Safe
V2	70	8	Warning

4.10 Performance Considerations

The system provides efficient performance in terms of communication range, power consumption, and response time. LoRa enables long-range communication with minimal energy usage, making it suitable for vehicular applications.

4.11 Advantages

- Long communication range
- Low power consumption
- Cost-effective solution
- Real-time warning system
- Reliable under different conditions

4.12 Limitations

- Limited data rate of LoRa
- GPS inaccuracies in dense environments
- Possible communication delay

CHAPTER 5

Results and Discussion

5.1 Introduction

In this chapter, the results of the implementation of the Lane Change Decision Aid and Warning System via LoRa-based Vehicle-to-Vehicle (V2V) communication are provided. The main goal of this chapter is to assess the performance of the proposed Lane Change Decision Aid and Warning System and its effectiveness in providing safe lane-change decisions for drivers.

5.2 Overview of Results

The performance of the proposed Lane Change Decision Aid and Warning System via LoRa-based Vehicle-to-Vehicle (V2V) communication is provided in this section. The proposed system is tested for different vehicle speeds and distances. The LoRa-based Vehicle-to-Vehicle (V2V) communication is working efficiently by transmitting data between the vehicles and sending warning messages based on safety conditions. The experimental results are provided in the following Table. The results show that when the distance

Table 5.1: Experimental Results

Vehicle ID	Speed (km/h)	Distance (m)	Status
V1	60	15	Safe
V2	70	8	Warning

between vehicles is less than the safe limit, the system generates a warning.

5.3 Analysis and Interpretation

Analysis of the system performance was performed according to the observed results. It is noted that the system is capable of detecting unsafe conditions.

The system is capable of transmitting data through the LoRa communication system, which is stable and covers long distances. The response time is also sufficiently quick for real-time systems. The relationship between the distance and the safety condition is as follows:

- If the distance is greater, then the condition is safe.
- If the distance is less, then the condition is warning.

It is noted that the system is functioning as expected.

5.4 Evaluation of Quality Factors

- **Environment:** The system is helpful in reducing accidents, which indirectly saves the environment from pollution and fuel consumption.
- **Sustainability:** The low-power communication makes the system sustainable.
- **Safety:** It enhances the safety of the driver by reducing the chances of accidents.
- **Ethics:** It does not involve any personal data and provides privacy to the user.
- **Cost:** It is cost-effective because the components used are low-cost.
- **Type:** It is an embedded system designed to operate in real-time.
- **Standards:** It follows the basic standards for communication and embedded systems.

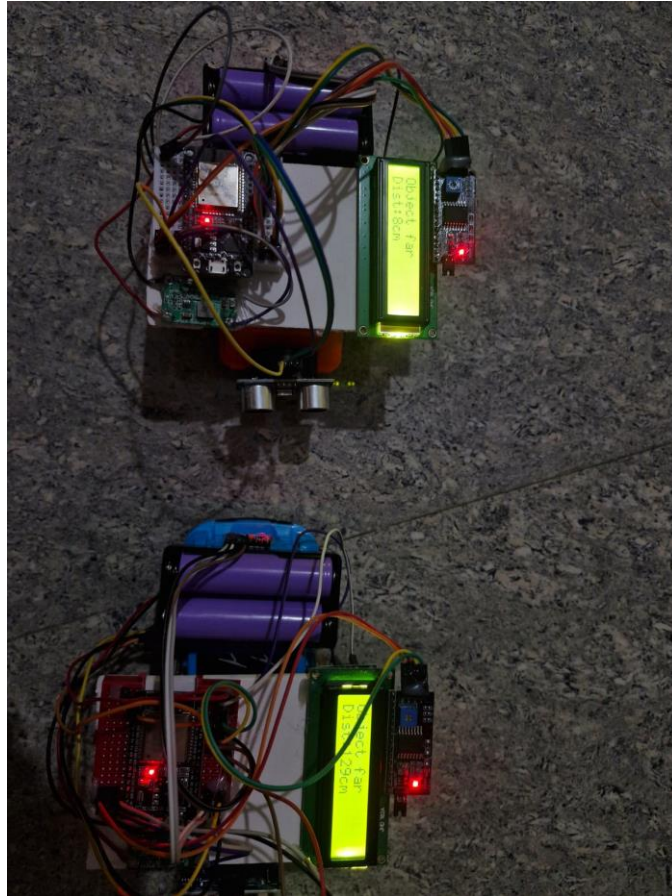


Figure 5.1: Prototype of Lane Change and Overtake Assistance System

5.5 Summary

The results show that the proposed Lane Change and Overtake Assistance System based on V2V communication using LoRa technology is efficient in recognizing unsafe road situations. The system is successful in recognizing nearby vehicles and lane positions in real time and thus can effectively detect possible collision situations during lane change and overtaking. The proposed system is successful in overcoming limitations associated with conventional sensor-based systems due to the implementation of LoRa technology in V2V communication. The implementation of sensor technology and image processing techniques has improved the accuracy of lane boundary detection. The proposed system has been successful in alerting the driver in time by displaying visual and audio signals, thus reducing response time and minimizing the possibility of road accidents.

The implementation of DSP technology has improved the quality of communication by reducing noise and efficiently transmitting signals. The experimental results show that the proposed system is successful in maintaining consistent performance in different road situations and has efficiently communicated between two vehicles. The proposed system has successfully demonstrated the feasibility of integrating LoRa technology with ITS. The proposed system is reliable, cost-effective, and efficient in road situations and thus can be effectively implemented in real-time situations. The proposed system is a cost-effective solution in improving road safety and can play a significant role in developing advanced driver assistance systems (ADAS) and autonomous vehicles.

CHAPTER 6

Conclusion and Future Scope

The system of Lane Change and Overtake Assistance is a reliable system for improving road safety by using LoRa-based V2V technology. There are various areas in which improvement can be made in the system. One area in which improvement can be made is by integrating 5G technology into the system. This can improve the rate of data transmission and make it reliable and faster. It can also enable ultra-low latency communication technology, which is highly useful in high-speed traffic situations. Another area in which improvement can be made is by extending the system by integrating cloud-based data analytics technology into the system. Another area where improvement can be made is in the creation of a user interface system where alerts and warnings can be shown to the driver. This will create a better understanding of the system for the driver. Another area where improvement can be made is in the integration of encryption techniques in the system to create secure communication technology that will protect data transmission from various cyber crimes. The efficiency of the system can also be increased through the creation of efficient low-power hardware modules, along with the application of intelligent power management techniques. This will ensure efficient working of the system. Furthermore, the system can be integrated with autonomous driving technologies. This will allow coordination between manual and autonomous driving, thus facilitating traffic flow.

CHAPTER 7

Mapping of Program Outcomes (POs), Program Specific Outcomes (PSOs), and Sustainable Development Goals (SDGs)

7.1 Mapping with Program Outcomes (POs)

The outcomes of the project are mapped with the Program Outcomes as follows:

- **PO1: Engineering Knowledge** – The knowledge of embedded systems, microcontrollers, and sensors used while developing the project.
- **PO2: Problem Analysis** – The real-time problems associated with lane change while driving, which can be solved by the embedded system.
- **PO3: Design/Development of Solutions** – The design and development of the project on an embedded system.
- **PO4: Conduct Investigations** – The testing and analysis of the project.
- **PO6: Engineer and Society** – The contribution of the project towards the safety of the public.
- **PO7: Environment and Sustainability** – The efficient use of the vehicle.
- **PO9: Individual and Team Work** – The effective team work involved while developing the project.
- **PO10: Communication** – The proper communication of the project ideas.

- **PO11: Project Management** – The proper management of the project.

7.2 Mapping with Program Specific Outcomes (PSOs)

PSO1: Embedded Systems

The project is related to embedded system design, microcontrollers, hardware components for implementing vehicle safety in real-time. Thus, it reflects the understanding of embedded systems in the context of vehicle safety.

7.3 Mapping with Sustainable Development Goals (SDGs)

The following are the sustainable development goals that can be achieved through this project:

- **SDG 9: Industry, Innovation, and Infrastructure** – It can help in the development of smart embedded systems in transportation.
- **SDG 11: Sustainable Cities and Communities** – It can help in developing smart transportation systems for the community.

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